

Containerized Interactive nnU-Net ResEnc for Whole-Body PET/CT Lesion Segmentation

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Abstract. We describe the AI4Viaevum submission to AutoPET V. Our entry is a faithful containerization of the official nnU-Net ResEnc interactive baseline for whole-body FDG PET/CT lesion segmentation, deployed under the inference contract of the challenge on grand-challenge.org. Rather than introducing a new architecture, this submission prioritizes *reproducible and contract-correct deployment*: we verify the full input/output interface, confirm offline (network-isolated) operation, and characterize the runtime envelope against the 15-minute-per-case limit of the challenge. We report what this submission is and, equally, what it is not, so that its position on the leaderboard is interpreted correctly.

1. Introduction

AutoPET V continues the AutoPET series on automated lesion segmentation in whole-body FDG PET/CT, adding an *interactive* setting in which sparse user clicks localize lesions of interest. Participants submit an algorithm container to grand-challenge.org, which executes it on a held-out test set under fixed compute and per-case time limits.

Our team entered this challenge late in its cycle. We therefore adopted an explicitly staged strategy: first establish a *correct, verified, submittable* container, and only then pursue methodological gains. This report documents the first stage, which is what the present submission contains.

2. Method

2.1 Segmentation backbone

We use the nnU-Net v2 framework with the residual-encoder (ResEnc) preset, which is the configuration distributed by the organizers as the interactive baseline. The self-configuring pipeline of nnU-Net derives patch size, spacing, normalization and network topology from a dataset fingerprint; the ResEnc variant replaces the plain encoder with residual blocks, which has been repeatedly shown to improve segmentation of small, scattered targets — the dominant failure mode in whole-body PET lesion segmentation, where lesions are numerous, small and distributed across the field of view.

We employ the officially released `fold_0` weights without additional training or fine-tuning. This is a deliberate choice given the time available, and it is stated plainly here because it determines how the resulting score should be read (Section 5).

2.2 Two-channel PET/CT input

PET (SUV) and CT volumes are supplied as separate channels. CT contributes anatomical context that disambiguates physiological uptake (brain, myocardium, bladder, renal collecting system) from pathological uptake; PET carries the metabolic signal. Both are resampled to the fingerprint spacing and normalized per the nnU-Net configuration.

2.3 Interactive click handling

The challenge supplies user interactions as `lesion-clicks.json`, in which each point carries a `name` of either `tumor` or `background`. Both polarities matter: foreground clicks nominate lesions, while background clicks are corrective, marking physiological uptake that should be suppressed.

Our pipeline converts the grand-challenge point format into the SW-FastEdit interaction representation, separating the two classes, and then encodes each class as a spatial click heatmap in the voxel coordinate frame of the volume. These heatmaps are appended to the PET and CT channels, so that the network receives interaction guidance in the same geometry as the image data. Cases with no clicks degrade gracefully to the fully automatic path rather than failing.

3. Implementation and Verification

3.1 Inference container

The algorithm is packaged as a Docker image conforming to the grand-challenge algorithm interface: inputs are read from the mounted input directory and the segmentation is written to the declared output socket. The image bundles the framework, the model weights, and all Python dependencies, with no runtime downloads.

Property	Value
Base framework	nnU-Net v2, ResEnc preset
Weights	Official AutoPET baseline, <code>fold_0</code> (checksum-verified)
Image size (compressed)	4.29 GB
Target instance	NVIDIA A10G, 24 GiB VRAM
Declared main memory	16 GB
Challenge time limit	15 min / case

3.2 Offline self-test

Prior to submission the container was executed with `--network=none` on a locally held case. We verified, in this isolated environment, that: (i) every Python import resolves without network access; (ii) the produced segmentation matches the expected geometry, spacing and label semantics; and (iii) the container exits cleanly with status 0.

This check is not a formality. In an earlier submission to a different MICCAI 2026 challenge, our team shipped a container carrying one unused import that failed only on the evaluation machine; a silent fallback path absorbed the exception and the algorithm returned degenerate output *while still producing a score*. Network-isolated import verification is now a mandatory gate for every container we submit, and we note it here because this failure mode is invisible on the leaderboard.

3.3 Platform validation

The uploaded image passed the grand-challenge import and activation check, confirming manifest validity, interface conformance and instance-type compatibility.

4. Results

At the time of writing, this submission is our first evaluated entry; official test-set metrics are therefore not yet available to us. We report the verification outcomes that were available before submission:

- Offline (`--network=none`) execution: **passed**, exit status 0
- Output geometry, spacing and label semantics: **conform** to the challenge contract
- Platform image import and activation: **passed**
- Per-case runtime under the A10G envelope: **within** the 15-minute limit

We deliberately do not quote self-computed accuracy figures, since we hold no labeled subset that is independent of the data the released weights were trained on; any such number would be optimistically biased and would misrepresent expected test-set behavior.

5. Discussion

What this submission is. A verified, contract-correct deployment of the official interactive baseline, establishing a reproducible floor for our team on this task.

What it is not. It is not a methodological contribution. It contains no architectural modification, no additional training, and no ensembling. Any leaderboard position it attains reflects the baseline of the organizers as executed by us, and should be read as such.

Intended next steps. The directions we consider most promising, in order: (i) fine-tuning on the AutoPET V training data rather than using released weights directly; (ii) multi-fold ensembling, which the 15-minute budget accommodates; (iii) an explicitly click-conditioned decoder, since the interactive signal is currently consumed only as an input channel; and (iv) suppression of physiological uptake through anatomical priors, which remains the dominant source of false positives in whole-body PET.

6. Conclusion

We present a verified containerization of the official nnU-Net ResEnc interactive baseline for AutoPET V. The contribution is one of engineering discipline rather than modeling: a container that provably runs offline, respects the declared interface, and fits the compute envelope. We report its derivative nature explicitly so that the submission is interpreted accurately.

References

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Use of AI assistance. An AI assistant was used for engineering work (container construction, verification tooling) and for drafting and language editing of this report. All technical claims were verified against actual execution logs, and the authors take full responsibility for the content.